

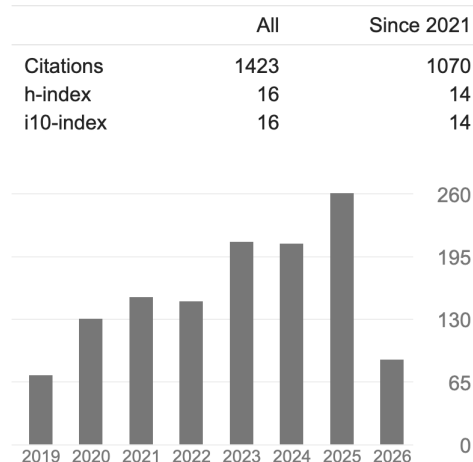
List of Publications

OVERVIEW

Total 1423 citations, h-index=16
(metrics by Google Scholar as of 16 Apr 2026)

Journal	Qty	Impact factor (2023)
Nature Photonics	1	39.728
Nature Communications	2	17.694
npj Quantum Information	1	10.758
Optica	3	10.644
IEEE TIFS	1	7.231
Physical Review X Quantum	1	7.514
Quantum	1	6.439
Phys. Rev. Applied	2	4.931
Phys. Rev. A	3	2.971
Physics Letters A	1	2.707
Journal Of Physics B	1	1.655
AQL Quantum	1	N/A
J. Surveill. Secur. Saf.	1	N/A
Total	19	

Cited by [VIEW ALL](#)



IN PREPARATION/UNDER REVIEW (4)

1. NQSN Singapore: A fully interoperable quantum-safe network testbed with versatile reference applications. In preparation.
H. Qin*, J. Y. Haw*, M. Wee, R. Frappier, C. Liang, X. Duan, Y. Cai, S. Sarda, K. Qiu, R. Murthy, T. Rimprongern, B. Sikdar, N. Ng, C. Kurtsiefer, M. Kasper, and A. Ling. (**first author, *contributed equally**)
2. A self-healing quantum network with all-transparent optical routers. In preparation.
H. S. Poh, X. Duan, M. Wee, Y. Cai, J. Y. Haw, H. Qin, M. Kasper, C. Kurtsiefer, and A. Ling
3. Analysis and Characterization Methodology of High Speed Balanced Homodyne Detector using RF MMIC Amplifier for Quantum Communication. In preparation.
4. RaspiQRNG: A do-it-yourself Quantum Random Number Generation on a Raspberry Pi. In preparation.

JOURNAL ARTICLES (19)

1. Fully passive quantum random number generation with untrusted light
APL Quantum 2 (4), 046105 (2025).
K. Qiu, Y. Cai, N. Ng and J. Y. Haw (**Last author**)
2. Integrated photonic platform with high-speed entanglement generation and witnessing.
Optica 12, 1737-1746 (2025).;
G. Zhang, C. Wang, K.T. Goh, S.Q. Ng, R. Ho, H. Semenenko, S.A. Srinivasan, H. Wang, Y. Chen, J. Y. Haw, X. Gong, J. V. Campenhout, C. C.-W. Lim.
3. Quantum-secured Data Centre Interconnect in a field environment
J. Surveill. Secur. Saf. (2024), 5, 184-97 (Invited submission, **corresponding author**)
K. Qiu, J. Y. Haw, H. Qin, N. H. Y. Ng, B. Sikdar, M. Kasper, A. Ling
4. Provably-secure quantum randomness expansion with uncharacterised homodyne detection
Nature Communications 14 (1), 316 (2023).
C. Wang, I. W. Primaatmaja, H. J. Ng, J. Y. Haw, R. Ho, J. Zhang, G. Zhang and C. C.-W. Lim
5. Discrete-variable quantum key distribution with homodyne detection
Quantum 6, 613 (2022).
I. W. Primaatmaja, C. Liang, G. Zhang, J. Y. Haw, C. Wang, and C. C.-W. Lim

6. Securing practical quantum cryptosystems with optical power limiters
PRX Quantum 2, 030304 (2021).
G. Zhang, I. W. Primaatmaja, J. Y. Haw, X. Gong, C. Wang, and C. C.-W. Lim
7. An integrated silicon photonic chip platform for continuous-variable quantum key distribution
Nature Photonics 13, 12 (2019).
G. Zhang, J. Y. Haw, H. Cai, F. Xu, S. M. Assad, J. F. Fitzsimons, X. Zhou, Y. Zhang, S. Yu, J. Wu, W. Ser, L. C. Kwek, and A. Q. Liu
8. Real-time source independent quantum random number generator with squeezed states
Phys. Rev. Applied 12, 034017 (2019).
T. Michel, J. Y. Haw, D. G. Marangon, O. Thearle, G. Vallone, P. Villoresi, P. K. Lam, and S. M. Assad
9. Entanglement properties of a measurement-based entanglement distillation experiment
Phys. Rev. A 99, 042304 (2019).
H. Jeng, S. Tserkis, J. Y. Haw, H. M. Chrzanowski, J. Janousek, T. C. Ralph, P. K. Lam, and S. M. Assad
10. Machine Learning Cryptanalysis of a Quantum Random Number Generator
IEEE Trans. on Information Forensics and Security 14, 403 (2019).
N. D. Truong*, J. Y. Haw*, S. M. Assad, P. K. Lam, and O. Kavehei (**first author, *contributed equally**)
11. Quantum enhancement of signal-to-noise ratio with a heralded linear amplifier
Optica 4, 1421 (2017).
J. Zhao, J. Dias, J. Y. Haw, T. Symul, M. Bradshaw, R. Blandino, T. Ralph, S. M. Assad, and P. K. Lam
12. Characterization of a measurement-based noiseless linear amplifier and its applications
Phys. Rev. A 96, 012319 (2017).
J. Zhao, J. Y. Haw, T. Symul, P. K. Lam, and S. M. Assad
13. Overarching framework between gaussian quantum discord and gaussian quantum illumination
Phys. Rev. A 95, 022333 (2017).
M. Bradshaw, S. M. Assad, J. Y. Haw, S.-H. Tan, P. K. Lam, and M. Gu
14. Surpassing the no-cloning limit with a heralded hybrid linear amplifier for coherent states
Nature Communications 7, 13222 (2016).
J. Y. Haw, J. Zhao, J. Dias, S. M. Assad, M. Bradshaw, R. Blandino, T. Symul, T. C. Ralph, and P. K. Lam (**first author**)
15. Experimental demonstration of Gaussian protocols for one-sided device-independent quantum key distribution
Optica 3, 634 (2016).
N. Walk, S. Hosseini, J. Geng, O. Thearle, J. Y. Haw, S. Armstrong, S. M. Assad, J. Janousek, T. C. Ralph, T. Symul, et al.
16. Replicating the benefits of Deutschian closed timelike curves without breaking causality
npj Quantum Information 1, 15007 (2015).
X. Yuan, S. M. Assad, J. Thompson, J. Y. Haw, V. Vedral, T. C. Ralph, P. K. Lam, C. Weedbrook, and M. Gu
17. Maximization of Extractable Randomness in a Quantum Random-Number Generator
Phys. Rev. Applied 3, 054004 (2015).
J. Y. Haw, S. Assad, A. Lance, N. Ng, V. Sharma, P. K. Lam, and T. Symul (**first author**)
18. Bridging the gap between the Jaynes-Cummings and Rabi models using an intermediate rotating wave approximation
Physics Letters A 379, 779 (2015).
Y. Wang and J. Y. Haw
19. Experimental verification of quantum discord in continuous-variable states
Journal of Physics B: Atomic, Molecular and Optical Physics 47, 025503 (2014).
S. Hosseini, S. Rahimi-Keshari, J. Y. Haw, S. M. Assad, H. M. Chrzanowski, J. Janousek, T. Symul, T. C. Ralph, and P. K. Lam

STANDARDISATION & TECHNICAL REPORTS (10)

1. Draft Reference Specification for Secure Deployment of QKD Networks (2025)
IMDA TSAC Focus Area (FA) 7 Secure Deployment of QKD Networks Task Force (TF) **(Editor)**
2. Study Report on QKD modules & networks evaluation & certification (2024)
IMDA TSAC Focus Area (FA) 7 Quantum Communication Networks Task Force (TF) **(Editor)**
3. [Reference Specification for Quantum Key Distribution Networks, \(2023\), Issue 1](#)
IMDA TSAC Focus Area (FA) 7 Quantum Communication Networks Task Force (TF) **(Editor)**
4. [A Study on Implementation Attacks against QKD Systems \(2023\)](#). Federal Office for Information Security, Germany. **(Expert Reviewer)**.
5. [Framework of quantum key distribution \(QKD\) protocols in QKD network \(2023\)](#)
International Telecommunication Union (ITU) Study Group 17 Question 15. **(Proposer & Contributor)**
6. Fourth JCA-QKDN Meeting **(Contributor)**
7. [ETSI GR QKD 017 Quantum Key Distribution \(QKD\); Network architectures](#)
ETSI Industry Specification Group QKD **(Contributor)**
8. [ETSI GS QKD 005 Quantum Key Distribution \(QKD\); Protocol structures and security proofs Revision](#)
ETSI Industry Specification Group QKD **(Contributor)**
9. [ETSI GS QKD 014 Quantum Key Distribution \(QKD\); Protocol and data format of REST-based key delivery API](#)
ETSI Industry Specification Group QKD **(Contributor)** ETSI Industry Specification Group QKD **(Contributor)**
10. [ETSI GS QKD 020 Quantum Key Distribution \(QKD\); Protocol and data format of REST-based Interoperable Key Management System API](#)
ETSI Industry Specification Group QKD **(Contributor)**

BOOKS & BOOK CHAPTERS (3)

1. "Sichere Zufallszahlengenerierung in Systemen mit kontinuierlichen Variablen."
[Quanten-Zufallszahlengenerierung: Theorie und Praxis.](#)
edited by C. Kollmitzer, S. Schauer, S. Rass, and B. Rainer, Cham: Springer International Publishing., (2024), (pp. 95-125).
[J. Y. Haw](#), S. M. Assad, and P. K. Lam **(Lead author)**
2. Secure Random Number Generation in Continuous Variable Systems
[Quantum Random Number Generation: Theory and Practice,](#)
edited by C. Kollmitzer, S. Schauer, S. Rass, and B. Rainer, Springer International Publishing, Cham, (2020), pp. 85 - 112.
[J. Y. Haw](#), S. M. Assad, and P. K. Lam **(Lead author)**
3. [Six Quantum Pieces: A first course in quantum physics.](#) World Scientific, (2010).
Scarani, Valerio, Chua Lynn, and Shiyang Liu & [J. Y. Haw*](#) **(Illustrator)**.

CONFERENCE PUBLICATIONS/PRESENTATIONS (35)

1. Development of National Quantum-Safe Network Testbed in Singapore
[H. Qin*](#), [J. Y. Haw*](#), T. Rimprongern, M. Wee, N. Ng, C. Kurtsiefer, M. Kasper, and A. Ling.
(first author, *contributed equally)
Quantum Engineering Sciences & Technologies for Industry & Services (2025)
2. Quantum-Secured Data Centre Interconnect in a field environment.
[K. Qiu](#), [J. Y. Haw](#), [H. Qin](#), N. H. Y. Ng, B. Sikdar, M. Kasper, A. Ling.
Institute Physics Singapore (IPS) Meeting (2025)
(Presenter, Best Poster Award)
3. Automated Switchover of Quantum Traffic with All-Transparent Optical Routers.
Institute Physics Singapore (IPS) Meeting (2025)
[H. S. Poh](#), [X. Duan](#), [M. Wee](#), [Y. Cai](#), [J. Y. Haw](#), [H. Qin](#), [C. Kurtsiefer](#), [M. Kasper](#), and [A. Ling](#)

4. NQSN Singapore: A fully interoperable quantum-safe network testbed with versatile reference applications.
Institute Physics Singapore (IPS) Meeting (2025)
H. Qin*, J. Y. Haw*, M. Wee, R. Frappier, C. Liang, X. Duan, Y. Cai, S. Sarda, K. Qiu, R. Murthy, T. Rimprongern, B. Sikdar, N. Ng, C. Kurtsiefer, M. Kasper, and A. Ling. (**first author, *contributed equally**)
5. NQSN Singapore: A fully interoperable quantum-safe network testbed with versatile reference applications.
Annual conference on quantum cryptography (QCrypt) (2025)
H. Qin*, J. Y. Haw*, M. Wee, R. Frappier, C. Liang, X. Duan, Y. Cai, S. Sarda, K. Qiu, R. Murthy, T. Rimprongern, B. Sikdar, N. Ng, C. Kurtsiefer, M. Kasper, and A. Ling. (**first author, *contributed equally**)
6. A self-healing quantum network with all-transparent optical routers
Quantum Innovation (2024)
X. Duan, H. S. Poh, M. Wee, Y. Cai, J. Y. Haw, H. Qin, M. Kasper, C. Kurtsiefer, and A. Ling
7. All-Transparent Optical Routers for a Quantum Network
Institute Physics Singapore (IPS) Meeting (2024)
H. S. Poh, X. Duan, M. Wee, Y. Cai, J. Y. Haw, H. Qin, C. Kurtsiefer, M. Kasper, and E. J. A. Ling
8. Secure Implementation and Verification of a Certifiable Source Device Independent Quantum Random Number Generator
Annual conference on quantum cryptography (QCrypt) (2024)
K. Qiu, Y. Cai, N. Ng and J. Y. Haw
9. The National Quantum-Safe Network in Singapore
Annual conference on quantum cryptography (QCrypt) (2024)
H. Qin*, J. Y. Haw*, M. Wee, C. Liang, X. Duan, Y. Cai, K. Qiu, S. Sarda, R. Murthy, R. Frappier, N. Ng, B. Sikdar, C. Kurtsiefer, M. Kasper and A. Ling (***contributed equally**)
10. Building a Fully Interoperable Quantum-Safe Network Testbed with Versatile Reference Applications
10th ETSI/IQC Quantum Safe Cryptography Conference (2024)
J. Y. Haw*, H. Qin*, X. Duan, Y. Cai, R. Murthy, M. Wee, C. Liang, K. Qiu, S. Sarda, N. Ng, B. Sikdar, C. Kurtsiefer, M. Kasper and A. Ling (**Presenting author, *contributed equally**)
11. On the certification of quantum random number generators
10th ETSI/IQC Quantum Safe Cryptography Conference (2024)
K. Qiu, Y. Cai, N. Ng and J. Y. Haw
12. Numerical Methods Evaluation for Key Rate Computation in Quantum Key Distribution
10th ETSI/IQC Quantum Safe Cryptography Conference (2024)
R. Chung, J. Y. Haw, N. Ng and Y. Cai
13. The National Quantum-Safe Network in Singapore
2023 European Conference on Optical Communications (ECOC) (2020)
H. Qin*, J. Y. Haw*, X. Duan, Y. Cai, R. Murthy, N. Ng, B. Sikdar, C. Kurtsiefer, M. Kasper and A. Ling (**Presenting author, *contributed equally**)
14. Quantum random number generation with uncharacterised homodyne detection
CLEO: Applications and Technology, JTh2A. 20 (2023)
C. Wang, I. W. Primaatmaja, H. J. Ng, J. Y. Haw, R. Ho, J. Zhang, G. Zhang, C. Wang, C. C.-W. Lim
15. Single-photon path entanglement based on integrated silicon photonics
2023 Conference on Lasers and Electro-Optics (CLEO). IEEE (2023)
G. Zhang, C. Wang, K. T. Goh, S. Q. Ng, R. Ho, H. Semenenko, S.A. Srinivasan, H. Wang, Y. Chen, J. Y. Haw, et al.
16. The National Quantum-Safe Network in Singapore
23rd Asian Quantum Information Science Conference (2023)
J. Y. Haw*, Hao Qin*, X. Duan, Y. Cai, R. Murthy, N. Ng, B. Sikdar, C. Kurtsiefer, M. Kasper and A. Ling (***Contributed equally**)

17. Building a certifiable source device independent quantum random number generator
23rd Asian Quantum Information Science Conference (2023)
K. Qiu, Y. Cai, N. Ng and J. Y. Haw
18. Integrated photonic platform with high-speed single-photon path entanglement
Annual conference on quantum cryptography (QCrypt) (2022)
G. Zhang, C. Wang, K. T. Goh, S. Q. Ng, H. Wang, Y. Chen, J. Y. Haw, X. Gong, C. C.-W. Lim
19. Quantum random number generation with uncharacterised homodyne detection
Annual conference on quantum cryptography (QCrypt) (2022)
C. Wang, I. W. Primaatmaja, J. Zhang, J. Y. Haw, R. Ho, G. Zhang, C. C.-W. Lim
20. Experimental proposal of discrete-variable quantum key distribution with homodyne detector
Annual conference on quantum cryptography (QCrypt) (2022)
C. C. Liang, C. Wang, I. W. Primaatmaja, G. Zhang, J. Y. Haw, C. C.-W. Lim
21. Analysis and Characterization Methodology of High Speed Balanced Homodyne Detector using RF MMIC Amplifier for Quantum Communication
Annual conference on quantum cryptography (QCrypt) (2022)
R. Ho, J. Y. Haw, X. Yan, Y. X. Guo, C. Wang, C. C.-W. Lim
22. Benchmarking a Quantum Random Number Generator with Machine Learning
Quantum Information and Measurement VI (2021), W3B. 5
J. Y. Haw, N.D. Truong, H.J. Ng, R. Ho, S. Assad, C. Wang, P.K. Lam, O. Kavehei
23. RaspiQRNG: A do-it-yourself Quantum Random Number Generation on a Raspberry Pi
2021 Institute of Physics Singapore (IPS) (2021)
J. Y. Haw, R. Ho, C. Liang, H. J. Ng, B. Goh, K. Kok, J. Yong, D. C. Wong, C. Wang, C. C.-W. Lim (**Presenting author**)
24. Optical Power Limiters for Securing Practical Quantum Communication Systems
Quantum Information and Measurement VI (2021), M2C. 6
G. Zhang, I. W. Primaatmaja, J. Y. Haw, X. Gong, C. Wang, C. C.-W. Lim
25. Discrete-Variable Quantum Key Distribution Based on Homodyne Detection
Quantum Information and Measurement VI (2021), M2C. 3
IW Primaatmaja, C. Liang, G Zhang, J. Y. Haw, C Wang, C. C.-W. Lim
26. Quantum Randomness Expansion with Uncharacterized Homodyne Detection Secure against Quantum Side-Information
Quantum Information and Measurement VI (2021), F2B. 7
I. W. Primaatmaja, J. Zhang, J. Y. Haw, R. Ho, G. Zhang, C. Wang, C. C.-W. Lim
27. Real-Time Self-Testing Quantum Random Number Generator with Non-classical States
Annual conference on quantum cryptography (QCrypt) (2020)
T. Michel, J. Y. Haw, D. G. Marangon, O. Thearle, G. Vallone, P. Villoresi, P. K. Lam, and S. M. Assad
28. Securing practical quantum cryptography with optical power limiters
Annual conference on quantum cryptography (QCrypt) (2020)
G. Zhang, I. W. Primaatmaja, J. Y. Haw, X. Gong, C. Wang, C. C.-W. Lim
29. Benchmarking a Quantum Random Number Generator with Machine Learning
Annual conference on quantum cryptography (QCrypt) (2020)
J. Y. Haw, N.D. Truong, H.J. Ng, R. Ho, S. Assad, C. Wang, P.K. Lam, O. Kavehei
30. Real-Time Self-Testing Quantum Random Number Generator with Non-Classical States
2020 European Conference on Optical Communications (ECOC) (2020)
T. Michel, J. Y. Haw, D. G. Marangon, O. Thearle, G. Vallone, P. Villoresi, P. K. Lam, and S. M. Assad (**Presenting author**)
31. Integrated chip for continuous-variable quantum key distribution using silicon photonic fabrication
2018 Conference on Lasers and Electro-Optics (CLEO) IEEE (2018)
G. Zhang, L. Cao, J. Y. Haw, X. Jia, X. Su, L. Kwek, J. Fitzsimons, P. Lam, X. Zhou, W. Gao, et al.

32. Quantum enhancement of signal-to-noise ratio for arbitrary coherent states using heralded linear amplifiers
2017 Conference on Lasers and Electro-Optics Pacific Rim (CLEO-PR) IEEE (2017)
J. Zhao, J. Dias, J. Y. Haw, M. Bradshaw, R. Blandino, T. Symul, T. Ralph, S. Assad, and P. K. Lam
33. Surpassing the no-cloning limit with a heralded hybrid linear amplifier
2017 Conference on Lasers and Electro-Optics Pacific Rim (CLEO-PR) IEEE (2017)
J. Y. Haw, J. Zhao, J. Dias, S. M. Assad, M. Bradshaw, R. Blandino, T. Symul, T. C. Ralph, and P. K. Lam (**Presenting author**)
34. Measurement-based noiseless linear amplification for quantum communication
Proc. SPIE, Quantum and Nonlinear Optics III, Vol. 9269 (2014).
H. M. Chrzanowski, N. Walk, J. Y. Haw, O. Thearle, S. M. Assad, J. Janousek, S. Hosseini, T. C. Ralph, T. Symul, and P. K. Lam
35. Experimental verification of quantum discord in continuous-variable states and operational significance of discord consumption
2014 Conference on Lasers and Electro-Optics (CLEO)-Laser Science to Photonic Applications (2014)
S. Hosseini, S. Rahimi-Keshari, J. Y. Haw, S. M. Assad, H. M. Chrzanowski, J. Janousek, T. Symul, T. C. Ralph, P. K. Lam, M. Gu, et al.

EXPERIMENTS SUPPORTED (NQSN TESTBED)

1. Metropolitan quantum key distribution using a Ga N-based room-temperature telecommunication single-photon source.
Physical Review Applied, 23(5), p.054022. (2025)
H. Zhang, et al.
2. Polarization-encoded quantum key distribution with a room-temperature telecom single-photon emitter.
National Science Review : nwaf147. (2025)
X. Zhang et al.
3. Demonstration of entanglement distribution over 155 km metropolitan fiber using a silicon nanophotonic chip.
Newton, 1, 10, 100303 (2025)
J. Du et al.
4. Fiber based interferometer for seismic detection.
Unconventional Imaging, Sensing, and Adaptive Optics Vol. 13149. SPIE, (2024).
J. Auccapuella et al.
5. Integrated polarization-entangled photon source for wavelength-multiplexed quantum networks
arXiv:2511.22680
X Shi et al.

Industry-link Research Projects

PROJECT 1: Provision of Quantum Consultancy and Support Services to IMDA

Funded under IMDA, the project objective is to advance quantum-safe technologies within the telecom sector. The intent is to spearhead research, development, and implementation strategies for quantum-safe solutions as a focus, including QKD and PQC technologies to ensure secure telecommunications against the backdrop of evolving quantum computing threats. The National Quantum-Safe Network Plus (NQSN+) is an IMDA initiative to support network operators to deploy quantum-safe networks nationwide, granting businesses easy access to quantum-safe solutions that safeguard their critical data. The scope would also entail the development, enhancement, and support of the NQSN, and identifying areas to strengthen for the NQSN+.

PROJECT 2: National-Quantum-Safe Network Testbed (2025-2029)

Funded under Quantum Engineering Programme (QEP) 3.0, NQSN testbed is enhancing its capabilities by integrating advanced detection equipment (e.g. SNSPDs), integration of an Optical Ground Station (OGS) to support satellite QKD, and expansion of the network nodes. The testbed also supports a broad range of research experiments to advance quantum communication R&D, while also contributing to the commercial pilot (NQSN+) and ongoing standardisation efforts across international bodies.

PROJECT 3: National-Quantum-Safe Network, NQSN (2021-2025)

Funded under Quantum Engineering Programme 2.0, this project aim to establish a national wide quantum-safe network testbed to demonstrate and evaluate the quantum-safe technologies with four pillars of activities: testbed, use cases & ecosystem standardization and quantum security lab with more than 30 local and international public/private organisations. Several achievements throughout the project are:

- Established a quantum-safe network with QKD and PQC technology
- Demonstrating more than 15 PoCs on QKD/PQC & QKD network (QKDN)
- Established the 1st local standard on QKD and proposed & led 1st international standard on QKD protocol framework in ITU
- Hosted & organized >10 international and local events
- 30+ Publications, proceedings, preprints & conferences

#	Project Partners	Summary
1	STT-GDC& IDQ	QKD feasibility in data center environment
2	SUTD & Thales	QKD integration with Thales and 5G testbed
3	AWS, Horizon Quantum Computing & Fortinet	QKD integration with Fortinet L3 encryptor with Edge Cloud Computing
4	Netlink Trust	QKD feasibility in GPON environment
5	Fortinet	QKD integration with Fortinet L3 encryptor/PQC
6	GovTech & ST Engineering	QKD integration with ST Engineering L2 encryptor; QKD link interoperability
7	evolutionQ	QKD integration with KMS for QKDN
8	Hitachi	QKD integration with Hitachi L2.5 encryptor
9	QuBridge	Long-range (cross-border) symmetric key exchange (plus QKD)
10	Terrabit-Ciena	QKD integration with Ciena L1 encryptor
11	Keysight	QKD link stability evaluation
12	Checkpoint, Fortinet	PQC integration in QKDN
13	Juniper	QKD integration with software L3 encryptor
14	Viavi	QKD/PQC evaluation with IPSec tunnel
15	SCION	Next generation software routing with PQC & QKD
16	S15 Instruments	QKD functional and security evaluation

Table 1: Industry-linked QKD and PQC research and deployments on NQSN Testbed, with fibre provisioning from Netlink Trust, Singapore.